

Proof. **(1) Degenerate and edge cases.**

(a) *All weight on a single index.* If $\lambda_k = 1$ and $\lambda_i = 0$ for $i \neq k$, then

$$f\left(\sum_{i=1}^n \lambda_i x_i\right) = f(x_k) = \sum_{i=1}^n \lambda_i f(x_i),$$

so equality holds.

(b) *Some weights equal zero.* If $\lambda_j = 0$ for some j , the term $\lambda_j x_j$ disappears from the weighted sum and $\lambda_j f(x_j)$ disappears from the right-hand side. Removing those terms yields an instance of (??) with fewer indices, so the inequality is unchanged.

(c) *The case $n = 1$.* When there is only one point, (??) reads $f(x_1) \leq f(x_1)$, which is true with equality.

Hence the theorem holds in all degenerate situations. In what follows we assume $n \geq 2$ and that at least two weights are positive.

(2) Base case $n = 2$. Let $\lambda_1 = t$, $\lambda_2 = 1 - t$ with $t \in [0, 1]$. The weighted average is $tx_1 + (1 - t)x_2$, and by convexity (??)

$$f(tx_1 + (1 - t)x_2) \leq tf(x_1) + (1 - t)f(x_2),$$

which is exactly (??) for $n = 2$. Thus Jensen's inequality holds for two points.

(3) Inductive step. Assume that (??) holds for any collection of $k \geq 2$ points with non-negative weights summing to one. We prove it for $n = k + 1$.

Let $\lambda_1, \dots, \lambda_{k+1}$ satisfy (??) and set

$$\mu := \sum_{i=1}^k \lambda_i \quad (0 \leq \mu \leq 1). \quad (4)$$

If $\mu = 0$, then $\lambda_{k+1} = 1$ and we are in the one-weight case already treated; hence we may assume $\mu > 0$.

Define the combined point

$$y := \frac{1}{\mu} \sum_{i=1}^k \lambda_i x_i. \quad (5)$$

Since the coefficients λ_i/μ are non-negative and sum to one, y is a convex combination of the first k points.

3.1 Two-point convexity applied to (y, x_{k+1}) . Using (??)–(??) we rewrite the full weighted average as

$$\sum_{i=1}^{k+1} \lambda_i x_i = \mu y + (1 - \mu)x_{k+1}. \quad (6)$$

Applying convexity (??) with $t = \mu$, $x = y$, $y = x_{k+1}$ gives

$$f(\mu y + (1 - \mu)x_{k+1}) \leq \mu f(y) + (1 - \mu)f(x_{k+1}). \quad (7)$$

3.2 Bounding $f(y)$ by the induction hypothesis. Since y is a convex combination of $x_1, \dots, x_k$, the induction hypothesis (with normalized weights λ_i/μ) yields

$$f(y) = f\left(\sum_{i=1}^k \frac{\lambda_i}{\mu} x_i\right) \leq \sum_{i=1}^k \frac{\lambda_i}{\mu} f(x_i). \quad (8)$$

Multiplying (??) by the positive μ we obtain

$$\mu f(y) \leq \sum_{i=1}^k \lambda_i f(x_i). \quad (9)$$

3.3 Completion of the inductive step. Insert (??) into (??) and use (??) for the argument of f :

$$f\left(\sum_{i=1}^{k+1} \lambda_i x_i\right) \leq \sum_{i=1}^k \lambda_i f(x_i) + (1 - \mu)f(x_{k+1}). \quad (10)$$

Because of (??), $1 - \mu = \lambda_{k+1}$, and (??) becomes

$$f\left(\sum_{i=1}^{k+1} \lambda_i x_i\right) \leq \sum_{i=1}^{k+1} \lambda_i f(x_i),$$

which is precisely (??) for $n = k + 1$.

Thus the inequality for k points implies the inequality for $k + 1$ points.

(4) Conclusion by induction. The base case $n = 2$ has been proved, and the inductive step shows that validity for k points implies validity for $k + 1$ points. By the principle of mathematical induction, Jensen's inequality

$$f\left(\sum_{i=1}^n \lambda_i x_i\right) \leq \sum_{i=1}^n \lambda_i f(x_i) \quad (3)$$

holds for every integer $n \geq 1$.

(5) Equality conditions.

- *General convex f .* Equality in (??) for a pair (x, y) and $t \in (0, 1)$ occurs iff f is affine on the interval $[x, y]$. In the inductive proof, equality in (??) forces f to be affine on the segment joining y and x_{k+1} , while equality in (??) forces f to be affine on $\text{conv}\{x_1, \dots, x_k\}$. Consequently, equality in (??) holds iff f is affine on the convex hull of $\{x_i : \lambda_i > 0\}$.

- *Strictly convex f .* If f is strictly convex, the only way an affine restriction can occur on a non-trivial interval is when that interval collapses to a point. Hence equality in (??) can occur only when all points with positive weight coincide, i.e. there exists $c \in \mathbb{R}$ such that $x_i = c$ whenever $\lambda_i > 0$.

In particular, equality also occurs when at most one weight λ_i is non-zero (the inequality reduces to an identity).

Therefore, Jensen's inequality holds for any convex function and any finite collection of real numbers with non-negative weights summing to one, with the equality cases described above. $\square$